

We could classify use cases into:

- **Core Clinical Use Cases** (business value)
- **Platform Use Cases** (supporting capabilities)

This makes the architecture and future documentation much more coherent.

Core Clinical Use Cases

These are the scenarios that hospital stakeholders actually care about.

UC1 – AI-Assisted Cardiac Segmentation

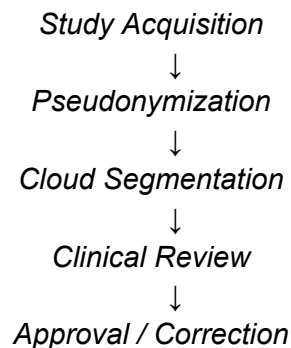
Business Goal

Automatically identify and segment cardiac structures from CT/MRI studies.

Actors

- Radiologist
- Cardiologist
- AI Segmentation Service

Typical Flow



Business Value

- Reduced manual segmentation effort
- Faster study preparation
- Standardized anatomical models

AI Components

- Computer Vision
- Deep Learning Segmentation Models

Human-in-the-Loop: Very High

UC2 – Automated 3D Cardiac Reconstruction

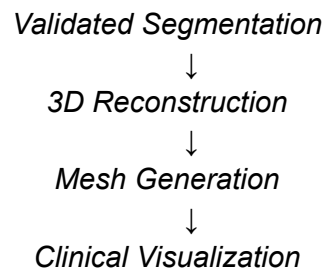
Business Goal

Generate navigable 3D heart models from segmented imaging studies.

Actors

- Cardiologist
- Surgeon
- 3D Reconstruction Engine

Typical Flow



Business Value

- Improved surgical planning
- Better anatomical understanding
- Enhanced clinician communication

AI Components

- Segmentation output consumption
- Geometry reconstruction algorithms

Human-in-the-Loop: Medium

UC3 – AI-Based Clinical Decision Support

Business Goal

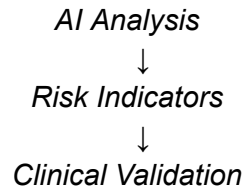
Assist clinicians in identifying potential anomalies and pathological patterns.

Actors

- Cardiologist
- Decision Support Engine

Typical Flow





Business Value

- Faster interpretation
- Improved diagnostic consistency
- Reduced oversight risk

AI Components

- Detection models
- Classification models
- Future LLM explanations

Human-in-the-Loop: Mandatory

UC4 – Virtual Heart Exploration

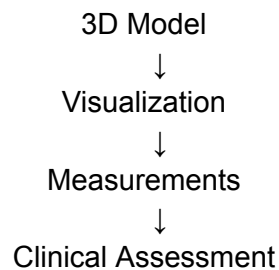
Business Goal

Enable interactive exploration of reconstructed anatomical models.

Actors

- Cardiologist
- Surgeon

Typical Flow



Business Value

- Improved planning
- Better communication
- Enhanced training

AI Components: Indirect

Human-in-the-Loop: Complete

Platform Use Cases

These are not directly visible to clinicians but are essential for the solution.

UC5 – Secure Imaging Acquisition & Pseudonymization

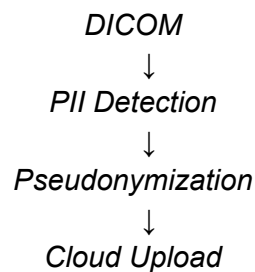
Goal

Acquire imaging studies while protecting patient identity.

Actors

- PACS
- Edge Platform

Flow



Business Value

- GDPR compliance
- Secure data sharing

UC6 – Human Validation & Feedback Collection

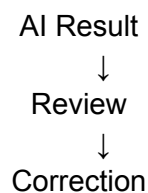
Goal

Capture clinician expertise and corrections.

Actors

- Clinician
- Feedback Service

Flow



↓
Feedback Repository

Business Value

- Quality assurance
- Ground truth generation

UC7 – Continuous Learning & Model Improvement

Goal

Improve AI models using validated feedback.

Actors

- MLOps Platform
- Data Scientists

Flow

Feedback
↓
Dataset Curation
↓
Retraining
↓
Model Promotion

Business Value

- Progressive accuracy improvements
- Long-term platform evolution

UC8 – Multi-Hospital AI Service Delivery

Goal

Serve multiple healthcare organizations from a centralized AI platform.

Actors

- Multiple Hospitals
- AWS Platform

Flow

Hospital A
Hospital B
Hospital C

↓
Shared AI Platform

Business Value

- Economies of scale
- Centralized governance
- Lower operational costs

StartUp Prioritization Proposed

Priority	Use Case	Reason
1	UC5 Secure Acquisition & Pseudonymization	Foundation of everything
2	UC1 AI-Assisted Segmentation	Core AI value proposition
3	UC6 Human Validation & Feedback	Key differentiator
4	UC7 Continuous Learning	Advanced AI architecture
5	UC2 3D Reconstruction	High business impact
6	UC3 Decision Support	Most complex AI use case
7	UC4 Virtual Exploration	Consumption layer
8	UC8 Multi-Hospital Delivery	Platform scaling